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React ES6 Arrow Functions

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Arrow Functions

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Arrow functions allow us to write shorter function syntax:

Example

Get your own React.js Server

Before:

```
hello = function() {
  return "Hello World!";
}
```

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```
hello = () => {
  return "Hello World!";
}
```

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It gets shorter! If the function has only one statement, and the statement returns a value, you can remove the brackets *and* the `return` keyword:

Example

Arrow Functions Return Value by Default:

```
hello = () => "Hello World!";
```

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Note: This works only if the function has only one statement.

If you have parameters, you pass them inside the parentheses:

Example

Arrow Function With Parameters:

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In fact, if you have only one parameter, you can skip the parentheses as well:

Example

Arrow Function Without Parentheses:

```
hello = val => "Hello " + val;
```

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What About `this`?

The handling of `this` is also different in arrow functions compared to regular functions.

In short, with arrow functions there is no binding of `this`.

In regular functions the `this` keyword represented the object that called the function, which could be the window, the document, a button or whatever.

With arrow functions, the `this` keyword *always* represents the object that defined the arrow function.

Let us take a look at two examples to understand the difference.

Both examples call a method twice, first when the page loads, and once again when the user clicks a button.

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Example

With a regular function, `this` represents the object that called the function:

```
class Header {
  constructor() {
    this.color = "Red";
  }

//Regular function:
  changeColor = function() {
    document.getElementById("demo").innerHTML += this;
  }
}

const myheader = new Header();

//The window object calls the function:
window.addEventListener("load", myheader.changeColor);

//A button object calls the function:
document.getElementById("btn").addEventListener("click", myheader.changeCol
```

Try it Yourself »

Example

With an arrow function, `this` represents the Header object no matter who called the function:

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```
  }

//Arrow function:
  changeColor = () => {
    document.getElementById("demo").innerHTML += this;
  }
}

const myheader = new Header();


//The window object calls the function:
window.addEventListener("load", myheader.changeColor);

//A button object calls the function:
document.getElementById("btn").addEventListener("click", myheader.changeCol
```

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Remember these differences when you are working with functions. Sometimes the behavior of regular functions is what you want, if not, use arrow functions.

Exercise ?

What is a correct arrow function syntax?

- `myarrow = function() { return 'Hello'; }`
- `myarrow = () => 'Hello';`
- `myarrow => 'Hello';`

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